

STAT 408: Introduction to Linear Models

Sections 500 and 502, Spring 2026

Section 500: TR, 11:10 am - 12:25 pm, BLOC 457

Section 502: TR, 09:35 am - 10:50 am, BLOC 457

This syllabus is tentative and subject to change: Any change will be announced in class and/or posted on the course website Canvas. Please check the course website regularly.

Instructor: Dr. Seongbaek Yi

- Office: 412 Blocker Building
- Office Hours: Tue & Wed, 4:00-5:30 pm, or by appointment.
- Email: sby0108@tamu.edu

Teaching Assistant

Section	500	502
Name & email	Kwonsang Jung kwons.jung@tamu.edu	Jifan Li (jifanli@exchange.tamu.edu)
Office Hours	4:00-6:00 pm, Thursday	3:00-5:00 pm, Monday
Office	BLOC 422	BLOC 406B

- Office Hours: Instructor and TA's will hold office hours in-person to answer any questions related to the course. The office hour schedule of TAs will be posted on Canvas once finalized. It is strongly recommended that you visit office hours whenever you have a question about the course.

Course Description:

- Introduction to the formulation of linear models and the estimation of the parameters of such models, with primary emphasis on least squares. Application to multiple regression and curve fitting.

Learning Objectives:

At the conclusion of this course, successful students will be able to:

- Explain the goal of regression analysis as discovering relationship among several variables. Use vectors and matrices in developing the least squares framework for regression analysis.
- Identify appropriate graphs, summary statistics, models, and inferential statistics used in regression analysis,
- Interpret graphs, statistics, and regression models in various contexts,
- Calculate summary and inferential statistics,
- Compare and contrast various models,
- Create appropriate models for various contexts,
- Infer appropriate conclusions about populations based on data,
- Explain and compare properties of summary and inferential statistics and models,
- Combine concepts in new ways to solve different statistical problems,
- Understand the need and check the assumptions of inference for linear models.

Prerequisites:

- Three semesters of calculus and MATH 304 (Linear Algebra and Matrices)
- A course in inferential statistics (STAT 211, 641, or 651)

Course Materials:

- Textbook: *A modern approach to regression with R*, Simon Sheather, 2009, Springer. This is an unlimited-user ebook available for free on or off campus. Thanks to the support of Texas A&M University Libraries, you can access it for free: <http://proxy.library.tamu.edu/login?url=https://dx.doi.org/10.1007/978-0-387-09608-7>
- Statistical Software R: We will be using the statistical software R for performing statistical analyses in this course. R is a free software that can either be downloaded onto your personal computer. Please ensure you can access and install R in your computer by the first week. R is freely available online. Instructions on obtaining R are given at Canvas under 'R

Help and Instructions', where you will also find useful links to manuals and tutorials on how to use R. You can also learn R from DataCamp.

- Note Regarding R: You will be ultimately responsible for: (a) ensuring you are comfortable with R and familiar with its basic usage, commands etc. (b) understanding all data analyses done via R throughout the course, including usage of all commands, packages etc., and (c) interpreting R output on exams.
- Canvas: Various information, including lecture slides, assignments, announcement and etc, will be uploaded in Canvas. Please check the Canvas regularly and make sure for the most updated information.

Course Schedule: All of these topics will be covered, but I may alter the order and length of time spent on any particular topic to accommodate this particular class of students. This schedule is presented here to give you an idea of the flow of the course and to identify which chapters in the text go with each topic. Future dates are tentative. Please RECHECK this schedule OFTEN so that you will be aware of any changes.

- Week 1: Syllabus, Review on Stat 211
- Week 2: Simple Linear Regression (SLR): Estimation (Ch2)
- Week 3: SLR: Inference (Ch2)
- Week 4: SLR: Testing (Ch2)
- Week 5: SLR: Diagnostics and Transformations (Ch3)
- Week 6: Weighted Least Squares (WLS) (Exam 1)
- Week 7: WLS, Review on Linear Algebra (Ch4)
- Week 8: Multiple Linear Regression (MLR) (Ch5)
- Week 9: No class (Spring Break)
- Week 10: MLR: Diagnostics (Ch6)
- Week 11: MLR: Transformations (Ch6) (Exam 2)
- Week 12: Variable Selection: Criteria, Methods (Ch7)
- Week 13: Logistic Regression: Single Predictor (Ch8)
- Week 14: Binary Logistic Regression (Ch8)
- Week 15: Serially Correlated Errors (Ch9)
- Week 16: Final Exams

Important Academic Dates

- 01/12: First Day of Classes
- 01/16: Last Day to Add/Drop Courses
- 02/17: Midterm 1
- 03/02: Mid-Semester Grades Due
- 03/9-3/13: Spring Break
- 03/24: Midterm 2
- 04/14: Q-Drop Deadline
- 04/28: Last Day of Classes
- 04/30: Final Exam. Sec. 502, 12:30-2:30 pm; Sec. 500, Sec. 500, 3:00-5:00 pm

Homework and Quizzes

- Homework will be posted in Canvas and Quizzes will be given during the class. The lowest of each part will be dropped when calculating your final grade.
- The quality of writing HW and logical presentation of the arguments leading to the solution, not just the correct answer, will contribute greatly to the HW grade.
- Homework has to be submitted electronically via Canvas as one single PDF file. Please scan your homework with pdf file and upload it. If your homework is unintelligible or difficult to read, marks will be deducted appropriately.
- If you miss the deadline for HW, you can still submit it by next day 11:59 pm, but 30% of your obtained scores will be deducted as a late-submission penalty. Each one-day delay counts the 20 % deduction of remaining scores.
- You may consult with other students about the homework, but always write up your solutions by yourself. You should never

just copy from another person. Do not include R programs and computer printouts in your HW, unless asked to do so.

- Missed assignments: Each homework must be turned in by 11:59 pm CST on the assigned due date. Late homework is accepted without an excuse that is recognized as valid by the university. Likewise, you will only be allowed to make up an exam if it is missed for a valid reason. See university rule for more details.

Grading Policy:

- Total grade consists of Midterm 1 (20 %), Midterm 2 (20%), Final (30%), HW (15%), and Quiz (15%).
- You will NOT be allowed any extra credit projects to compensate for a poor average. Everyone is given the same opportunity to do well in this class. The final course grade will be based on the standard scale: A: 90~100; B: 80~89; C: 70~79; D: 60~69; F: below 60.
- If the overall course evaluation participation rate in your section exceeds certain thresholds, each student in the section will receive bonus points as follows: 1 point for 55–69%, 2 points for 70–84% and 3 points for above 85%.
- Incomplete: A grade of “incomplete” is only an option for students who have completed more than 2/3 of the class (after the second exam) and have faced unexpected, uncontrollable, and debilitating personal circumstances.
- Attendance and participation during the lectures are encouraged and will be rewarded. They are integral parts of the learning process and greatly enhance the team work spirit which is much needed.

Makeup Policy:

- If you missed a homework assignment or exam, see the university rule on Attendance website Rule 7: <http://student-rules.tamu.edu/rule07>
- If you fail to submit a homework assignment by the due date because of a university excused absence or due to illness or circumstances beyond your control, notify me in writing or by email (before, if feasible, otherwise within two working days after you return). If your absence is approved, I will notify you on how you may make up the missed assignment.
- If you must miss an exam because of a university excused absence or due to illness or circumstances beyond your control, notify me in writing or by email (before, if feasible, otherwise within two working days after you return). If your absence is approved, I will notify you on how you may make up the exam.
- If you miss a homework assignment or an exam and your reason for missing the assignment or exam is not accepted, then you will receive a score of 0 for the assignment or exam.
- A temporary grade of I (Incomplete) at the end of a semester indicates that the student has completed the course with the exception of a major quiz, final exam, or other work. The instructor shall give this grade only when the deficiency is due to an authorized absence or other cause beyond the control of the student.

Communication:

- All questions about course materials should be addressed during office hours or during lecture.
- Please use email only for administrative issues. In all your emails to me and the TA, please include the prefix “[STAT 408]” in your subject line headers (or else, your email may go unanswered!).

Acceptable Uses of Generative Artificial Intelligence:

ChatGPT and other generative AI are freely available tools that can perform a variety of functions for us. However, it's important to understand how such tools are allowed to be used in this course.

Acceptable uses of generative AI in this course include:

- Editing or rephrasing written work that has already been written by the student to improve the syntax, grammar and overall readability of the work.
- Synthesizing or explaining course concepts while learning and studying to contribute to their understanding of the course material
- Looking up appropriate syntax of individual R functions for use in a data analysis or for understanding errors that may arise when running R code.

However, the work turned in by students must ultimately be their own and students will therefore be accountable for the work they turn in. Unacceptable uses of generative AI in this course include:

- Copying from any generative artificial intelligence applications, including ChatGPT and other AI writing and coding assistants, for the purpose of completing assignments in this course.
- Producing an entire data analysis, written report, or any other piece of work meant for grades.

In summary, generative AI like ChatGPT can be really helpful in your learning process and to improve skills valued in the workplace. However, it cannot be used as a substitute for learning and material produced from these tools should not be passed off as your own. This would be considered academic misconduct. The instructor therefore reserves the right to ask students to explain their work and their process for creating their assignment in an oral examination.

Academic Integrity Statement and Policy:

“An Aggie does not lie, cheat or steal, or tolerate those who do.”

“Texas A&M University students are responsible for authenticating all work submitted to an instructor. If asked, students must be able to produce proof that the item submitted is indeed the work of that student. Students must keep appropriate records at all times. The inability to authenticate one’s work, should the instructor request it, may be sufficient grounds to initiate an academic misconduct case” (Section 20.1.2.3, Student Rule 20).

You can learn more about the Aggie Honor System Office Rules and Procedures, academic integrity, and your rights and responsibilities at aggiehonor.tamu.edu.

Statement on Plagiarism: The handouts used in this course are copyrighted. As such, you do not have the right to copy them unless I expressly grant permission. As commonly defined, plagiarism consists of passing off as one’s own ideas, words, writing, etc., that belong to another. In accordance with this definition, you are committing plagiarism if you copy the work of another person and turn it in as your own, even if you have the permission of that person. If you have any questions, please consult the latest issue of the Texas A&M University Student Rules, under the section “Scholastic Dishonesty.”

Copyright Statement: Faculty members own copyright in their educational work at Texas A&M University, as stated in the Texas A&M University System Policy for Intellectual Property Management and Commercialization. Students are not allowed to post or share any materials created by a faculty member unless given permission by that faculty member. This includes but is not limited to homework assignments, homework solutions, exams, exam solutions, lecture notes and any other supplemental materials. Any violation of this copyright policy could result in disciplinary actions as described in Student Rule 20.2: Procedures in Scholastic Dishonesty Cases and Student Rule 20.1.2.3.1

Cheating: Intentionally using or attempting to use unauthorized materials, information, notes, study aids or other devices or materials in any academic exercise. Unauthorized materials may include anything or anyone that gives a student assistance and has not been specifically approved in advance by the instructor. Texas A&M complies fully with the Digital Millennium Copyright Act (“DMCA”). Users of the Texas A&M network found to have engaged in repeated infringement of copyright are subject to termination of their network access and may be reported to the appropriate Dean or Human Resources officer for disciplinary action. Please see TAMU’s Copyright Infringement Policies and Sanctions Notification for additional information.

Americans with Disabilities Act (ADA) Policy: Texas A&M University is committed to providing equitable access to learning opportunities for all students. If you experience barriers to your education due to a disability or think you may have a disability, please contact Disability Resources in the Student Services Building or at (979) 845-1637 or visit disability.tamu.edu. Disabilities may include, but are not limited to attentional, learning, mental health, sensory, physical, or chronic health conditions. All students are encouraged to discuss their disability related needs with Disability Resources and their instructors as soon as possible.

Title IX and Statement on Limits to Confidentiality: Texas A&M University is committed to fostering a learning environment that is safe and productive for all. University policies and federal and state laws prohibit gender-based discrimination and sexual harassment, including sexual assault, sexual exploitation, domestic violence, dating violence, and stalking.

With the exception of some medical and mental health providers, all university employees (including full and part-time faculty, staff, paid graduate assistants, student workers, etc.) are Mandatory Reporters and must report to the Title IX Office if the employee experiences, observes, or becomes aware of an incident that meets the following conditions (see University Rule

08.01.01.M1):

- The incident is reasonably believed to be discrimination or harassment.
- The incident is alleged to have been committed by or against a person who, at the time of the incident, was (1) a student enrolled at the University or (2) an employee of the University.

Mandatory Reporters must file a report regardless of how the information comes to their attention – including but not limited to face-to-face conversations, a written class assignment or paper, class discussion, email, text, or social media post. Although Mandatory Reporters must file a report, in most instances, you will be able to control how the report is handled, including whether or not to pursue a formal investigation. The University's goal is to make sure you are aware of the range of options available to you and to ensure access to the resources you need.

Students wishing to discuss concerns in a confidential setting are encouraged to make an appointment with Counseling and Psychological Services (CAPS).

Students can learn more about filing a report, accessing supportive resources, and navigating the Title IX investigation and resolution process on the University's Title IX webpage.

Statement on Mental Health and Wellness: Texas A&M University recognizes that mental health and wellness are critical factors that influence a student's academic success and overall wellbeing. Students are encouraged to engage in proper self-care by utilizing the resources and services available from Counseling & Psychological Services (CAPS). Students who need someone to talk to can call the TAMU Helpline (979- 845-2700) from 4:00 p.m. to 8:00 a.m. weekdays and 24 hours on weekends. 24-hour emergency help is also available through the National Suicide Prevention Hotline (800-273-8255) or at suicidepreventionlifeline.org.

COVID: To help protect Aggieland and stop the spread of COVID-19, Texas A&M University urges students to be vaccinated and to wear masks in classrooms and all other academic facilities on campus, including labs. Doing so exemplifies the Aggie Core Values of respect, leadership, integrity, and selfless service by putting community concerns above individual preferences. COVID-19 vaccines and masking — regardless of vaccination status — have been shown to be safe and effective at reducing spread to others, infection, hospitalization, and death.